

1 Write a Python program using functions and recursion to calculate the factorial of a number.

AIM

To write a Python program that uses functions and recursion to calculate the factorial of a given number.

ALGORITHM

1. Start the program.
2. Define a recursive function :
 - If $n == 0$ or $n == 1$, return 1 (base case).
 - Otherwise, return $n * \text{factorial}(n - 1)$ (recursive case).
3. In the main program, take input from the user.
4. Call the function and display the result.
5. End program.

CODE

```
# Recursive function to calculate factorial
def factorial(n):
    if n == 0 or n == 1: # base case
        return 1
    else:
        return n * factorial(n - 1) # recursive case

# Main Program
num = int(input("Enter a number: "))

print(f"Factorial of {num} is {factorial(num)}")
```

OUTPUT

Enter a number: 5

Factorial of 5 is 120

CONCLUSION

This program demonstrates:

- Functions (encapsulates logic).
- Recursion (function calls itself until base case).
- Conditional Statements (base case check).
- A real-life application of recursion in mathematics and computer science.

2 Write a program to find the sum of digits of a number using recursion.

AIM

To write a Python program that uses functions and recursion to calculate the sum of digits of a given number.

ALGORITHM

1. Start the program.
2. Define a recursive function :
 - If , return (base case).
 - Otherwise, return (recursive case).
3. In the main program, take input from the user.
4. Call the function and display the result.
5. End program.

CODE

Recursive function to calculate sum of digits

```
def sum_of_digits(n):
```

```
    if n == 0: # base case
```

```
        return 0
```

else:

 return (n % 10) + sum_of_digits(n // 10) # recursive case

Main Program

num = int(input("Enter a number: "))

print(f"Sum of digits of {num} is {sum_of_digits(num)}")

OUTPUT

Enter a number: 1234

Sum of digits of 1234 is 10

CONCLUSION

This program demonstrates:

- Functions (encapsulates logic).
- Recursion (function calls itself until base case).
- Mathematical operations (for last digit, for removing last digit).